



Spring 2026 MEMS 3430 Design of Thermal Systems

Design Homework 1: Brayton Topping Cycle

Professor: Dr. Weisensee

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I hereby certify that the lab report herein was completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies. I acknowledge that artificial intelligence (AI) tools were used to generate content included in this report.

Samantha Feldman

A handwritten signature in black ink that reads "Samantha Feldman".

Assumptions and Boundary Conditions

Assumptions

- Assume ideal gas and air standard properties according to the problem statement; Therefore c_p is constant.
- Air is the working fluid of the open Brayton cycle according to the problem statement.
- The topping cycle is an open system. The working fluid of air is taken in and exhausted into the ambient atmosphere. Therefore, mass is crossing the boundary into and out of the system.
- Constant mass flow rate throughout the system due to the principle of conservation of mass in an ideal and air standard Brayton cycle.
- Kinetic and potential energy terms are negligible due to such small changes in velocity and elevation, and steady flow.
- In the solar tower there is perfect solar-to-thermal conversion. All solar energy gets converted into heat, without consideration of angle effects.
- Power plant location: Southern California Desert Region, specifically the Mojave Desert [1]. The Mojave Desert is popular for solar power generation as it has intense consistent sunlight for the majority of the year with little cloud coverage.

Boundary Conditions

- Isentropic processes between states: $1 \rightarrow 2s$, and $3 \rightarrow 4s$.
- Isobaric processes between states: $2s$, $2 \rightarrow 3$, and $4s$, $4 \rightarrow 5 \rightarrow 1$. There is no significant pressure difference across heat exchangers.
- Adiabatic processes between states: $1 \rightarrow 2$, and $3 \rightarrow 4$.
- No work done between states: $2 \rightarrow 3$, and $4 \rightarrow 5 \rightarrow 1$.
- $T_1 = 300$ K (entering the combustor), $T_3 = 1200$ K (leaving the heater), and $T_5 = 500$ K (exiting the topping cycle).

- Output heat transfer is fixed at 30 MW.
- The compression ratio range of $r = p_2/p_1$ is from 5 to 30.
- Daily solar irradiation value in the Mojave Desert = $6.0 \text{ kWh/m}^2/\text{day}$ [2].

Additional Constants

- $c_p = 1.004 \text{ kJ/kg-K}$.
- $k = 1.400$.

Brayton Cycle Diagrams

Figure 1 displays the schematic of a Combined Cycle Gas Turbine (CCGT) [3]. In this design homework, only the topping cycle is being analyzed.

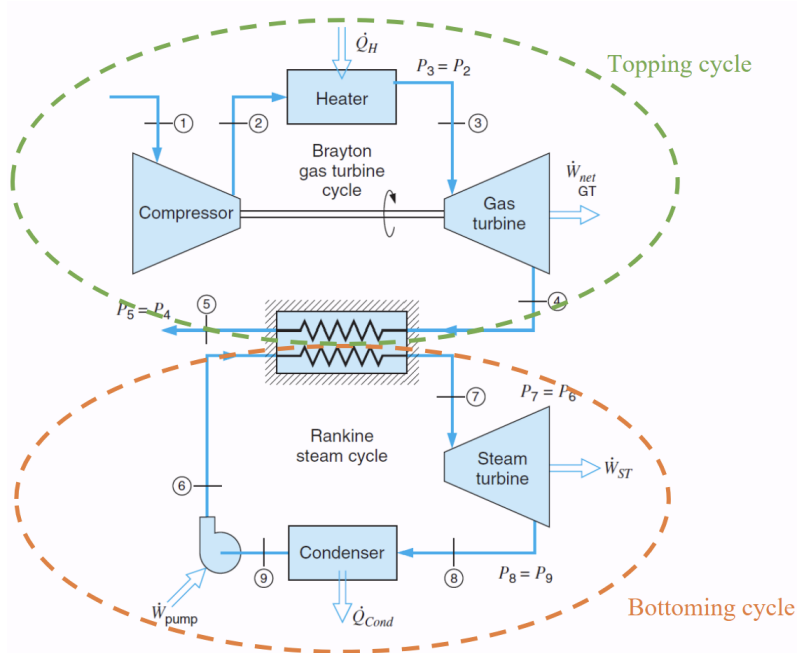


Figure 1 Schematic of a Combined Cycle Gas Turbine (CCGT) [3].

Figure 2 shows a graphical representation of the relationship between temperature and entropy for the 5-stage Brayton cycle. The state numbers correspond to the state numbers displayed in Fig 1.

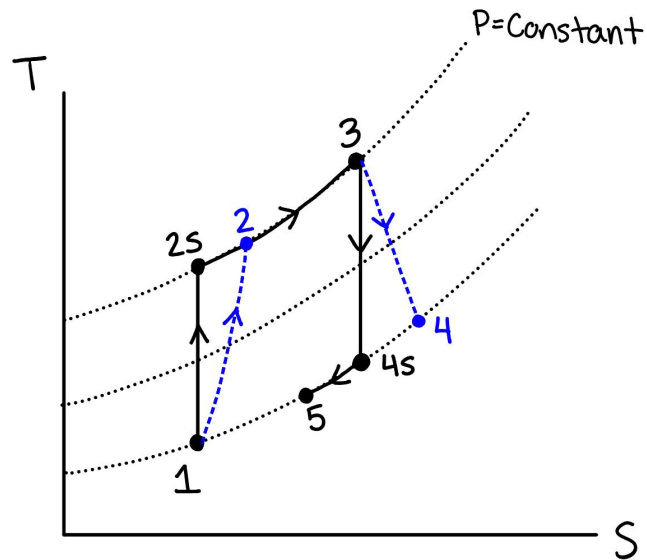


Figure 2 T-S Diagram of the Brayton Cycle.

Figure 3 shows a graphical representation of the relationship between pressure and volume for the 5-stage Brayton cycle.

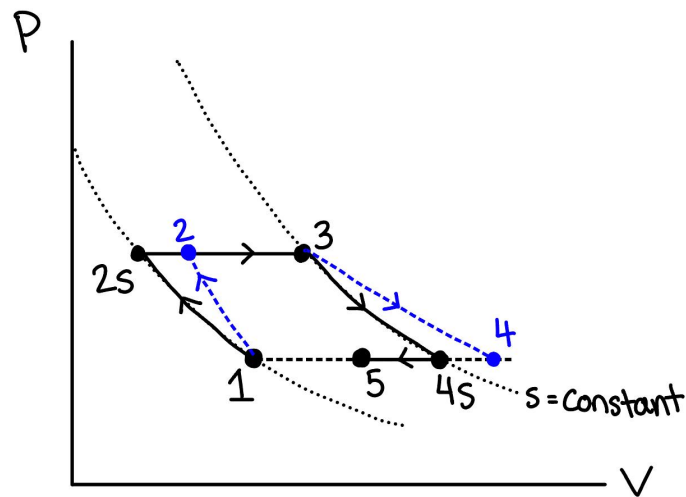


Figure 3 P-V Diagram of the Brayton Cycle.

Equations

To fully define the net power output and the thermal efficiency of the topping cycle, many undefined variables must be first be isolated and defined in relation to boundary conditions. After these variables are defined, Matlab was utilized to run the labeled equations and plot the net power output vs. compression ratio and thermal efficiency vs compression ratio for seven turbine and compressor isentropic efficiency combinations.

Finding Unknown Temperatures

Polytropic relationships [4]:

Utilizing the isentropic processes between $1 \rightarrow 2s$, and $3 \rightarrow 4s$, the ideal temperatures T_{2s} and T_{4s} were isolated and calculated.

$$r = \frac{P_2}{P_1} = \left(\frac{T_{2s}}{T_1} \right)^{\frac{k}{k-1}}$$
$$T_{2s} = T_1 (r)^{\frac{k-1}{k}} \quad (1)$$

$$1/r = \frac{P_4}{P_3} = \left(\frac{T_{4s}}{T_3} \right)^{\frac{k}{k-1}}$$
$$T_{4s} = T_3 (1/r)^{\frac{k-1}{k}} \quad (2)$$

Where T_1 is the real temperature [K] entering the compressor. T_{2s} is the ideal and T_2 is the real temperature [K] leaving the compressor. T_3 is the real temperature [K] entering the turbine. T_{4s} is the ideal and T_4 is the real temperature [K] leaving the turbine. The variable r is the pressure ratio and $k = C_p/C_v$. C_p is the coefficient of pressure [kJ/kg-K] and C_v is the coefficient of volume [kJ/kg-K].

Efficiencies [4]:

The relationship between the real and ideal temperatures in the isentropic efficiency equations for both the turbine and compressor were used to isolate and find the real T_2 and T_4 temperatures using Eq. 1 and 2.

$$\eta_t = real/ideal$$

$$\eta_t = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{T_3 - T_4}{T_3 - T_{4s}}$$

$$T_4 = T_3 - \eta_t(T_3 - T_{4s}) \quad (3)$$

$$\eta_c = ideal/real$$

$$\eta_c = \frac{h_1 - h_{2s}}{h_1 - h_2} = \frac{T_1 - T_{2s}}{T_1 - T_2}$$

$$T_2 = T_1 - \frac{T_1 - T_{2s}}{\eta_c} \quad (4)$$

Where η_t is the turbine isentropic efficiency and η_c is the compressor isentropic efficiency. The letter h denotes the enthalpy per unit of mass [kJ/kg] at each respective location notated by the subscript number that corresponds to the state in the schematic in Fig 1.

Finding Mass Flow Rate and Work

Energy Balances [4]:

By isolating the low-temperature heat exchanger an expression of the mass flow rate is calculated with an energy balance for the control volume of the heat exchanger using Eq. 3.

$$\dot{m} * h_4 = \dot{Q}_L + \dot{m} * h_5$$

$$\dot{m} * c_p(T_4 - T_5) = \dot{Q}_L$$

$$\dot{m} = \frac{\dot{Q}_L}{C_p(T_4 - T_5)} \quad (5)$$

Where $\dot{m}$ is the mass flow rate [kg/s], T_4 is the temperature [K] going into the heat exchanger, and T_5 is the temperature [K] exiting the heat exchanger. $\dot{Q}_L$ is the thermal energy [MW] given off by the low-temperature heat exchanger to the bottoming cycle.

The turbine was then isolated and an energy balance employed to craft an expression for the outgoing turbine power using Eq. 3 and 5.

$$\dot{m} * h_3 = \dot{W}_t + \dot{m} * h_4$$

$$\dot{W}_t = \dot{m} c_p(T_3 - T_4) \quad (6)$$

Where $\dot{W}_t$ is the power [W] output by the turbine.

Isolating the compressor and performing an energy balance determines an expression for the input compressor power using Eq. 4 and 5.

$$\dot{m} * h_1 + \dot{W}_c = \dot{m} * h_2$$

$$\dot{W}_c = \dot{m} c_p (T_2 - T_1) \quad (7)$$

Where $\dot{W}_c$ is the power [W] input by the compressor.

To form the final expression for net power the power being input must be subtracted from the outgoing power by implementing Eq. 6 and 7.

$$\dot{W}_{net} = \dot{W}_t - \dot{W}_c$$

$$\dot{W}_{net} = \dot{m} c_p (T_3 - T_4 - T_2 - T_1) \quad (8)$$

Where $\dot{W}_{net}$ is the net work [W] done by the system.

Finding Heat Input and Thermal Efficiency

To find the thermal efficiency, first the input thermal energy must be found. The input thermal energy was determined through an energy balance in which the heater was isolated with Eq. 4 and 5.

$$\dot{Q}_h + \dot{m} h_2 = \dot{m} h_3$$

$$\dot{Q}_h = \dot{m} c_p (T_3 - T_2) \quad (9)$$

Where $\dot{Q}_h$ is the input thermal energy [W].

Now with an expression for net power and input thermal energy, the thermal efficiency is able to be calculated with the expression below by inputting Eq. 8 and 9.

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_h} \quad (10)$$

Where η_{th} is the thermal efficiency.

Isentropic Efficiency Combination Plots

Figure 4 shows the plot of the thermodynamic cycle efficiency vs the compression ratio for the topping Brayton cycle.

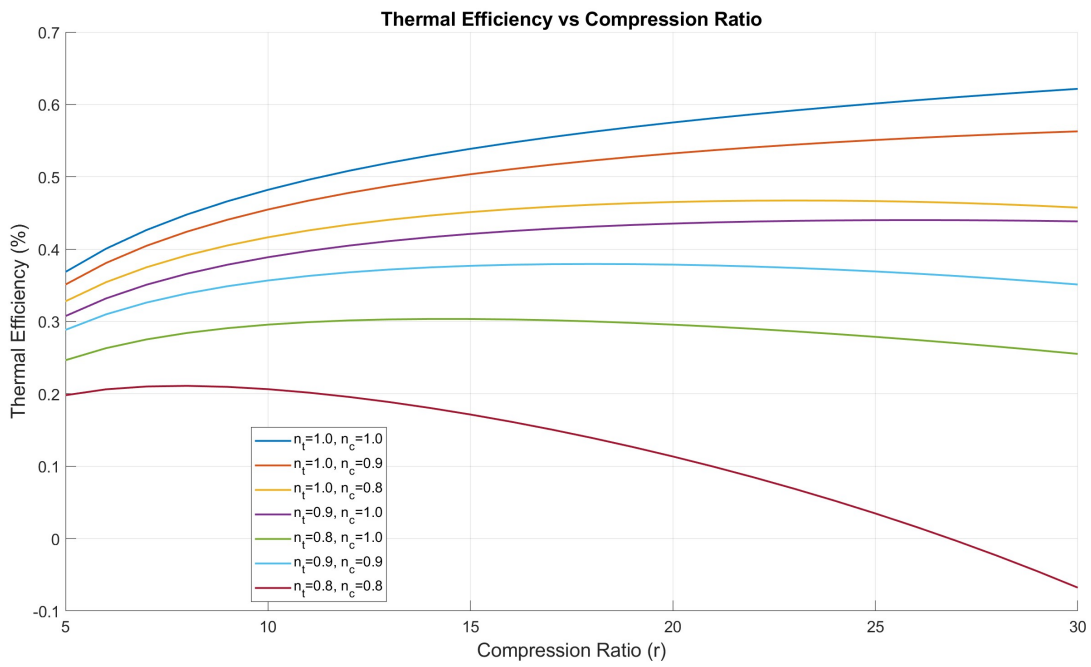


Figure 4 Plot of thermodynamic cycle efficiency vs. compression ratio.

Figure 5 shows the plot of the net power output vs the compression ratio for the topping Brayton cycle.

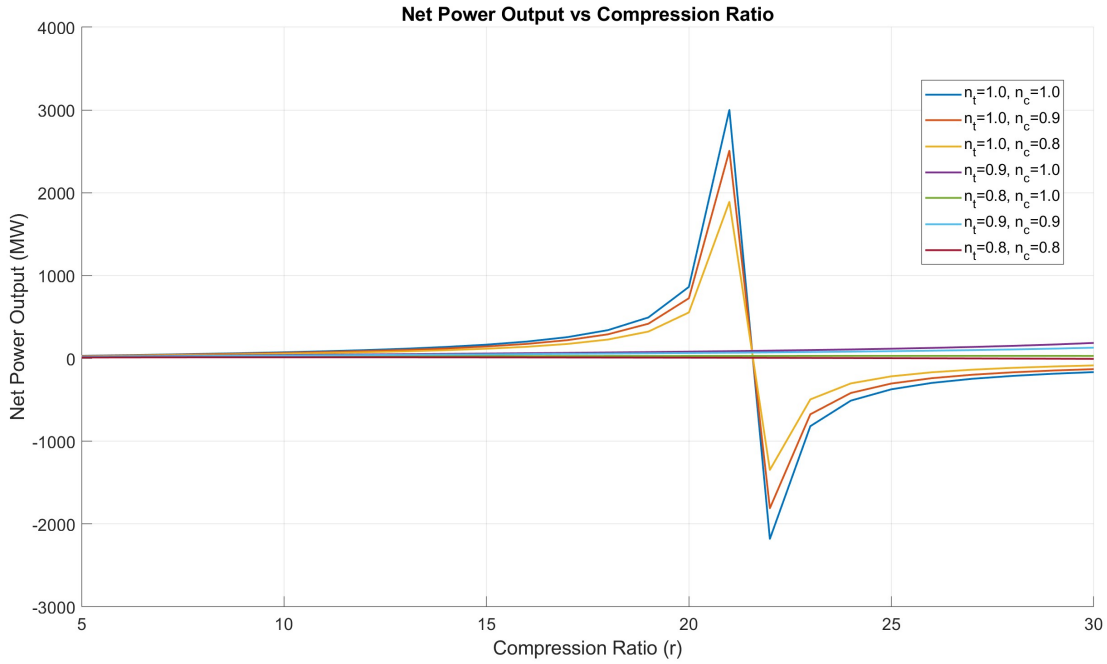


Figure 5 Plot of net power output vs. compression ratio.

Discontinuity Explanation

A discontinuity appears in the net power plot for the cases with turbine isentropic efficiency equal to 1 because these cases represent the ideal upper limit of turbine performance. As the pressure ratio increases, the turbine exhaust temperature T_4 decreases, which reduces the temperature difference across the low-temperature heat exchanger, as described by Eqs. 2 and 3. As this temperature difference decreases, the required mass flow rate increases. When $T_4 = T_5$ the temperature driving force across the heat exchanger vanishes, and the mass flow rate expression becomes proportional to $\dot{Q}_L/0$, rendering it undefined. This occurs at a pressure ratio of approximately 21.6 and produces the observed discontinuity in the net power plot. Beyond this point, $T_4 > T_5$, the heat exchanger would reject heat rather than absorb it, and the model predicts an asymptotic approach to zero net work.

From a macroscopic energy balance perspective, the turbine work and compressor work approach one another as the pressure ratio increases. The discontinuity corresponds to the point at which the work required by the compressor equals the work produced by the turbine, resulting

in zero net work output. Beyond this point, the compressor work exceeds the turbine work, so the cycle would require net work input rather than producing power, causing the net power to become negative. This behavior appears as a discontinuity because the mathematical model transitions from a power-producing Brayton cycle to an operating regime that is no longer physically viable. Once the turbine is already ideal, further increases in pressure ratio cannot recover positive net work, since the additional compression work grows faster than the additional expansion work.

This behavior is not observed when the compressor isentropic efficiency is equal to 1 because an ideal compressor only reduces the required compression work and does not impose an upper limit on the turbine work. Even with a perfect compressor, the turbine remains non-ideal in these cases, so the turbine work does not reach the same limiting balance with the compressor work. As a result, the net work varies smoothly with pressure ratio and does not exhibit the same abrupt discontinuity.

Compression Ratios that Optimize the Cycle Efficiency

Table 2 displays the compression ratios that optimize the cycle efficiency for each isentropic efficiency combination.

Table 1 Optimized compression ratios for each isentropic efficiency combination

Combination (η_t, η_c)	Optimal Compression Ratio
(1.0, 1.0)	21
(1.0, 0.9)	21
(1.0, 0.8)	21
(0.9, 1.0)	30
(0.8, 1.0)	14
(0.9, 0.9)	18
(0.8, 0.8)	8

Equipment Improvement Recommendation

The component that researchers and engineers should prioritize for performance improvements is the turbine. As shown in Fig. 4, the turbine has a stronger influence on the overall thermodynamic efficiency of the cycle than the compressor. All curves corresponding to the highest thermal efficiencies

are associated with the highest turbine isentropic efficiency, regardless of the compressor isentropic efficiency. Conversely, the isentropic-efficiency pair curves show a consistent decrease in thermal efficiency as turbine efficiency decreases, with only a weak dependence on compressor efficiency.

Solar Power Tower

Figure 6 displays the plot of thermodynamic cycle efficiency vs compression ratio for a realistic combination set of compressor and turbine isentropic efficiencies.

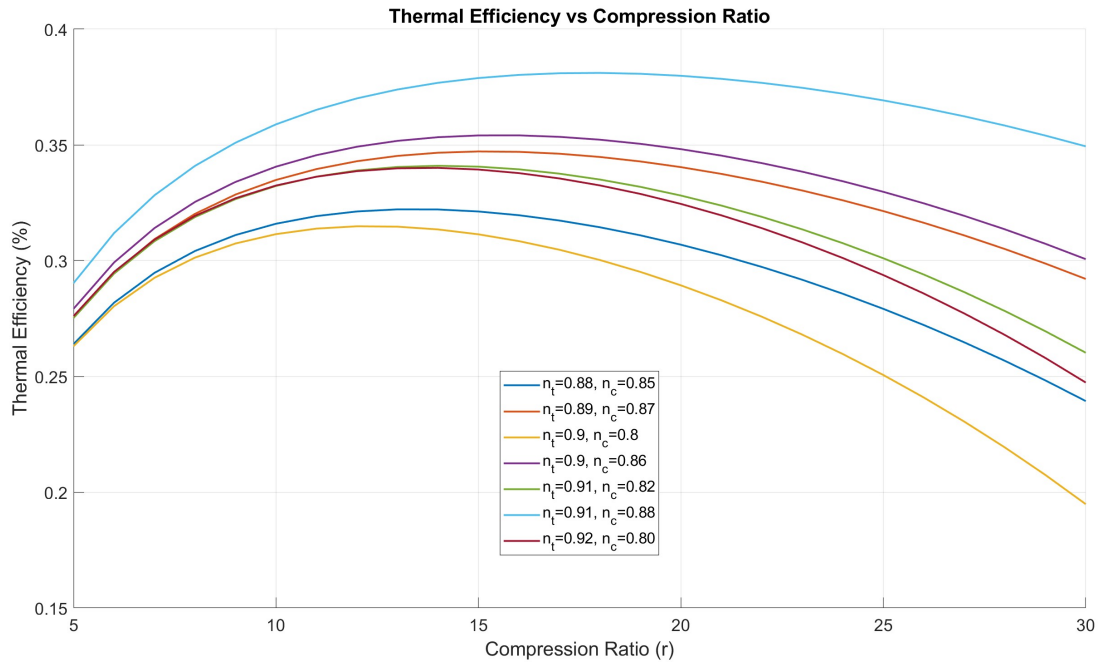


Figure 6 Plot of thermodynamic cycle efficiency vs. compression ratio for a set of realistic combinations of isentropic efficiencies.

Figure 7 displays the plot of net power output vs compression ratios for a realistic combination set of compressor and turbine isentropic efficiencies.

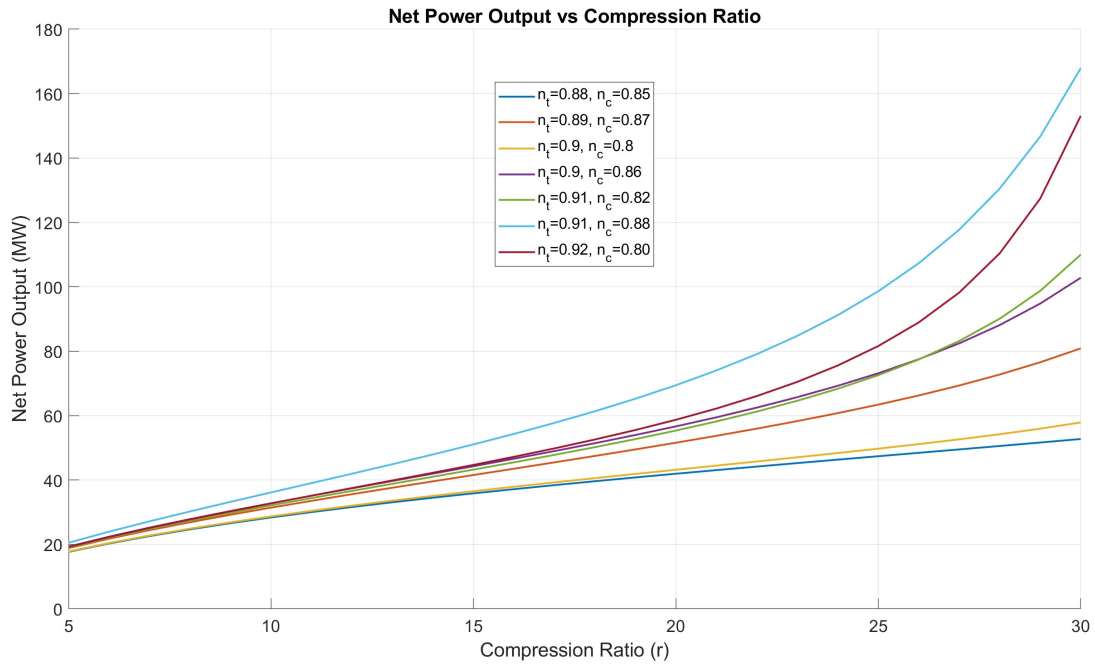


Figure 7 Plot of net power output vs. compression ratio for a set of realistic combinations of isentropic efficiencies.

Table 2 displays the compression ratios that optimize the cycle efficiency for each isentropic efficiency combination.

Table 2 Optimized compression ratios and corresponding necessary heat input for each realistic isentropic efficiency combination

Realistic Combination (η_t, η_c)	Optimal Compression Ratio	Necessary Heat Input (MW)
(0.88, 0.85) [5]	13	102.70
(0.89, 0.87) [6]	15	119.68
(0.9, 0.8) [5]	12	101.78
(0.9, 0.86) [5]	16	131.6
(0.91, 0.82) [7]	14	120.3
(0.91, 0.88) [8]	18	160.99
(0.92, 0.8) [8]	14	124.24

The efficiency pair that requires the lowest necessary heat input of 101.78 MW and has the lowest optimal compression ratio of 12 is the $\eta_t = 0.9$ and $\eta_c = 0.8$ pairing. Therefore, this pairing will be used as the pair of isentropic efficiencies to power the solar power tower.

The location of the solar power tower is in the southern California desert region, specifically in the Mojave Desert [1]. The daily solar irradiation value in the Mojave Desert is $6.0 \text{ kWh/m}^2/\text{day}$ [2]. The equation for power for a solar power tower can be used to isolate area and find the necessary number of 10 m by 10 m heliostats necessary to produce the amount of power needed.

$$P = A * I * \eta \quad (11)$$

Where P is power [W], A is area [m^2], I is daily solar irradiation value (DNI) in [W/m^2], and η is the reflectivity efficiency. With the assumption of perfect solar to thermal conversion regardless of consideration of angle effects η is 100 percent or 1. To convert the given DNI value into usable units a unit conversion is done as shown with the equation below.

$$6\text{kWh/m}^2/\text{day} * 1\text{day}/24\text{hours} * 1000\text{W}/1\text{kW} = 250\text{W/m}^2$$

Now isolating the area and plugging in the power, DNI, and efficiency values solves for the necessary area the heliostats must cover.

$$A = \frac{P}{I * \eta} \quad (12)$$

$$A = \frac{101.78 * 10^6}{250 * 1) = 407,120m^2$$

To find the necessary number of heliostats, the total necessary area is divided by the area of an individual heliostat.

$$407,120m^2 / 100m^2 = 4071.2$$

There cannot be 0.2 of a heliostat, so to produce the necessary heat input of 101.78 MW, the solar power tower would need 4072 heliostats of a 10m x 10m size.

Discussion and Sanity Check

Examining the optimal compression ratios corresponding to the seven isentropic efficiency combinations, the cases with a turbine isentropic efficiency of 1 all yield an optimal compression ratio of 21. This value corresponds to the condition where the turbine work equals the compressor work for an ideal turbine, which leads to the observed discontinuity in the net power output plot. In addition, the compression ratio of 30 obtained for the combination $(\eta_t, \eta_c) = (0.9, \text{ and } 1.0)$ is similarly at the high end and is typically only encountered in very large industrial or specialized power-generation applications [5]. At such high compression ratios, the compressor blade heights would become very small, which increases aerodynamic losses and would likely reduce the compressor isentropic efficiency, rendering these operating points mechanically and aerodynamically impractical. In contrast, the less ideal and more realistic efficiency combinations yield optimal compression ratios in the range of approximately 8 to 18, which is consistent with values typically encountered in practical gas turbine systems [8].

To produce an exit temperature for the topping cycle of 500 Kelvin at an efficiency combination of $\eta_t = 90\%$ and $\eta_c = 80\%$, the optimal compression ratio of 12 is reasonable, as it provides sufficient turbine expansion to reduce the turbine inlet temperature to the desired exhaust level without requiring an extreme pressure ratio. Additionally, the required heat input of 101.78 MW is of a reasonable

magnitude because producing a net power output of 32.05 MW from this input corresponds to a thermal efficiency of 31.48%, which is consistent with expectations for a non-ideal Brayton cycle with $\eta_t = 90\%$ and $\eta_c = 80\%$ and is of the correct order for a realistic Brayton-cycle topping system.

In the extremely sunny location of the Mojave Desert, the DNI value is $6.0 \text{ kWh/m}^2/\text{day}$ [2]. This value was used to estimate the number of heliostats (mirrors of size 10 m by 10 m) required to concentrate solar energy and provide the necessary thermal input of 101.78 MW. The resulting estimate is 4072 heliostats with the assumption of zero thermal loss. While this is a large number of mirrors, it is reasonable given the high thermal power demand. For reference, this corresponds to a total reflective area of approximately $4.07 * 10^5 \text{ m}^2$, which is consistent with the scale of existing solar-thermal power tower facilities [1]. Together, these comparisons confirm that the predicted pressure ratios, power levels, and collector scale are physically reasonable and consistent with real-world systems.

References

- [1] 2025, "Ivanpah Solar Electric Generating System," California Energy Commission, accessed January 31, 2026, <https://www.energy.ca.gov/powerplant/solar-thermal/ivanpah-solar-electric-generating-system>
- [2] Roberts, B. J., 2018, "Global Horizontal Solar Irradiance," National Renewable Energy Laboratory, accessed January 31, 2026, https://www.nrel.gov/docs/libraries/gis/high-res-images/nsrdb-v3-ghi-2018-01.jpg?sfvrsn=855ad6e1_1
- [3] Weisensee, P., 2026, "MEMS 3430 - Design of Thermal Systems Design Homework 1: Brayton Topping Cycle," Washington University in St. Louis.
- [4] Weisensee, P., 2026, "MEMS 412: Design of Thermal Systems Equation and Tables Package," Washington University in St. Louis.
- [5] Zivic, M., Galovic, A., and Virag, Z., 2014, "Detailed Analysis of the Effect of the Effect of the Turbine and Compressor Isentropic Efficiency on the Thermal and Energy Efficiency of a Brayton Cycle," Thermal Science, accessed February 2, 2026, <https://thermalscience.rs/pdfs/papers-2014/TSCI1403843Z.pdf>
- [6] Saravanamuttoo, H., Rogers, G., Cohen, H., Straznicky, P., and Nix, A., 2018, "Gas Turbine Theory," Pearson, accessed February 2, 2026, <https://www.pearson.com/en-gb/subject-catalog/p/gas-turbine-theory/P200000005323/9781292093130?srsId=AfmBOorCr2rdFcOYIUyRHPtSwW9DcEP1BiS1R9dygXHr1yavGuSv4iMc>
- [7] Ridwan, M. and Wibisono, F., 2024, "Analysis of Brayton Cycle Efficiency at 100 MW Load Before and After Overhaul of Unit 1 Block 1 at PLTGU PT. PLN Nusantara Power Up Gresik," Asian Journal of Social and Humanities, accessed February 2, 2026, <https://pdfs.semanticscholar.org/74f1/7c17f54d73fc1926ebf4ea179d4516020bf3.pdf>
- [8] Stewart, W. L., Benser, W. A., Glassman, A. J., Guentert, D. C., and Hickel, R. O., 1968, "Brayton Cycle Systems," NASA, accessed February 2, 2026, <https://ntrs.nasa.gov/api/citations/19690003247/downloads/19690003247.pdf>

A Acknowledgment of Artificial Intelligence Resources

AI tools were used to look up LaTeX and Matlab code and was verified by inputting into Overleaf and Matlab respectively. Additionally, AI tools were used as a starting point for research on the topic of realistic efficiencies, all information included from the output was verified by consulting the source. Anything not verified was not included in the report. A list of the AI prompts used is included below.

Cited AI Prompts

- ChatGPT, response to “matlab syntax to make an array of values that range from 5 to 30” OpenAI, January 19, 2026.
- ChatGPT, response to “matlab syntax to plot seven isentropic efficiency combinations with compressor efficiency denoted as η_c and turbine efficiency denoted as η_t . The first plot being thermodynamic cycle efficiency as the y axis vs compression ratio (r) as the x axis. ” OpenAI, January 19, 2026.
- Perplexity.ai, response to “give me a list of a realistic combination of compressor and turbine efficiencies and the link to the source they came from. Give them in the format: [turbine isentropic efficiency, compressor isentropic efficiency]. They all must be from a Brayton gas turbine cycle with the working fluid being air” Perplexity, January 21, 2026.
- Perplexity.ai, response to “I need a replacement for the combinations that came from reddit and turbomachinerymag.com, replace them with combinations from reliable sources” Perplexity, January 21, 2026.
- Gemini, response to “realistic combinations of compressor and turbine isentropic efficiencies examples and trustworthy source links” Google, January 21, 2026.
- Copilot, response to “How do I get this matlab code to print 1 QH for each combination of compressor and expander efficiencies” Microsoft, January 27, 2026.
- Gemini, response to “how to make an italic heading in overleaf” Google, February 1, 2026.
- ChatGPT, response to “any information to add to make this engineering sanity check more complete:” OpenAI, February 4, 2026.

B MATLAB Code

Part A: Given Isentropic Efficiency Combination Plots

```
1 % Constants
2 t1 = 300;
3 t3 = 1200;
4 t5 = 500;
5 qL = 30e6; % 30 MW
6 cp = 1.004e3; % J/(kg*K)
7
8 % Compression ratio range
9 r = 5:30;
10
11 effs = [ ...
12     1.0 1.0;
13     1.0 0.9;
14     1.0 0.8;
15     0.9 1.0;
16     0.8 1.0;
17     0.9 0.9;
18     0.8 0.8 ];
19
20 labels = { ...
21     'n_t=1.0, n_c=1.0'
22     'n_t=1.0, n_c=0.9'
23     'n_t=1.0, n_c=0.8'
24     'n_t=0.9, n_c=1.0'
25     'n_t=0.8, n_c=1.0'
26     'n_t=0.9, n_c=0.9'
27     'n_t=0.8, n_c=0.8'};
28
29 figure
30 hold on
31 grid on
32
33 for i = 1:size(effs,1)
```

```

34     nt = effs(i,1);
35     nc = effs(i,2);
36
37     t2s = t1 .* r.^(0.4/1.4);
38     t4s = t3 .* (1./r).^(0.4/1.4);
39
40     t2 = t1 + ((t2s - t1)./nc);
41     t4 = t3 - (nt.*(t3 - t4s));
42
43     mdot = qL./(cp .* (t4 - t5));
44     wt = mdot .* cp .* (t3 - t4);
45     wc = mdot .* cp .* (t2 - t1);
46     wnet = wt - wc;
47     qH = mdot .* cp .* (t3 - t2);
48
49     n_th = wnet ./ qH;
50
51     plot(r, n_th, 'LineWidth', 1.5)
52 end
53
54 xlabel('Compression Ratio (r)')
55 ylabel('Thermal Efficiency (%)')
56 title('Thermal Efficiency vs Compression Ratio')
57 legend(labels, 'Location', 'best')
58
59
60
61 figure
62 hold on
63 grid on
64
65 for i = 1:size(effs,1)
66     nt = effs(i,1);
67     nc = effs(i,2);
68
69     t2s = t1 .* r.^(0.4/1.4);
70     t4s = t3 .* (1./r).^(0.4/1.4);
71

```

```

72     t2 = t1 + (t2s - t1)./nc;
73     t4 = t3 - nt.*(t3 - t4s);
74
75     mdot = qL ./ (cp .* (t4 - t5));
76     wt = mdot .* cp .* (t3 - t4);
77     wc = mdot .* cp .* (t2 - t1);
78     wnet = wt - wc;
79
80     plot(r, wnet/1e6, 'LineWidth', 1.5) % GW
81 end
82
83 xlabel('Compression Ratio (r)')
84 ylabel('Net Power Output (MW)')
85 title('Net Power Output vs Compression Ratio')
86 legend(labels, 'Location', 'best')

```

Part B: Solar Power Tower Isentropic Efficiency Plots and Necessary Heat Input

```

1  % Actual values from literature search
2
3  % Constants
4  t1 = 300;
5  t3 = 1200;
6  t5 = 500;
7  qL = 30e6; % 30 MW
8  cp = 1.004e3; % J/(kg*K)
9
10 % Compression ratio range
11 r = 5:30;
12
13 effs2 = [ ...
14     .88 .85;
15     .89 0.87;
16     .9 0.8;
17     0.9 .86;
18     .91 .82;
19     .91 .88;
20     0.92 0.8 ];

```

```

21
22 labels = { ...
23     'n_t=0.88, n_c=0.85'
24     'n_t=0.89, n_c=0.87'
25     'n_t=0.9, n_c=0.8'
26     'n_t=0.9, n_c=0.86'
27     'n_t=0.91, n_c=0.82'
28     'n_t=0.91, n_c=0.88'
29     'n_t=0.92, n_c=0.80'};
30
31
32 figure
33 hold on
34 grid on
35
36 for i = 1:size(effs2,1)
37     nt = effs2(i,1);
38     nc = effs2(i,2);
39
40     t2s = t1 .* r.^(0.4/1.4);
41     t4s = t3 .* (1./r).^(0.4/1.4);
42
43     t2 = t1 + ((t2s - t1)./nc);
44     t4 = t3 - (nt.*(t3 - t4s));
45
46     mdot = qL./(cp .* (t4 - t5));
47     wt = mdot .* cp .* (t3 - t4);
48     wc = mdot .* cp .* (t2 - t1);
49     wnet = wt - wc;
50     qH = mdot .* cp .* (t3 - t2);
51
52     n_th = wnet ./ qH;
53     %QH_list(i) = mean(qH);
54     fprintf('QH for nt=%.2f, nc=%.2f:\n', nt, nc);
55     disp(qH)
56
57     plot(r, n_th, 'LineWidth', 1.5)
58 end

```

```

59
60 %disp(QH_list); %MW
61 xlabel('Compression Ratio (r)')
62 ylabel('Thermal Efficiency (%)')
63 title('Thermal Efficiency vs Compression Ratio')
64 legend(labels, 'Location', 'best')
65
66
67
68 figure
69 hold on
70 grid on
71
72 for i = 1:size(effs2,1)
73     nt = effs2(i,1);
74     nc = effs2(i,2);
75
76     t2s = t1 .* r.^(0.4/1.4);
77     t4s = t3 .* (1./r).^(0.4/1.4);
78
79     t2 = t1 + (t2s - t1)./nc;
80     t4 = t3 - nt.*(t3 - t4s);
81
82     mdot = qL ./ (cp .* (t4 - t5));
83     wt = mdot .* cp .* (t3 - t4);
84     wc = mdot .* cp .* (t2 - t1);
85     wnet = wt - wc;
86
87     plot(r, wnet/1e6, 'LineWidth', 1.5) % GW
88 end
89
90 xlabel('Compression Ratio (r)')
91 ylabel('Net Power Output (MW)')
92 title('Net Power Output vs Compression Ratio')
93 legend(labels, 'Location', 'best')

```